Lab Session 08

Open Ended Lab

OBJECTIVE

The objective of this lab is to evaluate students' understanding and practical skills developed during the previous seven lab sessions. Students will apply concepts such as software documentation, UML modeling, design principles, design patterns, and user interface prototyping to analyze and design a small software system. The lab allows students to creatively apply the knowledge gained in earlier labs to a problem of their choice.

EVALUATION METHOD

Students will be evaluated through open-ended problem-solving tasks. Each student will select a **small software system of their choice** and perform basic design and documentation activities using the tools and techniques learned in previous lab sessions.

Evaluation will be carried out using the **OEL rubrics (5 marks)** where each criterion will be assessed as **Achieved (1)** or **Not Achieved (0)**.

TASKS

System Selection

Students will select any small software system of their choice with at least one user and a few basic features.

Task 1 – Software Description

Write a short description of the selected system using **LaTeX**. The description should include:

- System name
- Purpose of the system
- Main features of the system
- Types of users (if applicable)

Task 2 – UML Modeling

Create two UML diagram for the selected system using any diagramming tool. Students may choose any one of the following diagrams:

- Use Case Diagram
- Class Diagram
- Sequence Diagram

- Collaboration Diagram

The diagram should clearly represent the structure or interaction of the system.

Task 3 – User Interface Prototype

Design a basic user interface prototype of the system using Figma. The prototype should include at least two screens, such as:

- Login or Home screen
- Main feature screen (e.g., task list, booking page, quiz page, etc.)

The design should follow basic **UI design principles**.

Task 4 – Design Concept Identification

Identify one design principle or design pattern (e.g., Singleton, Observer, or Facade) that could be applied in the selected system and briefly explain its possible use.

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Department of Software Engineering
OEL RUBRICS
Course Code and Title

Criteria	Achieved (1)	Not Achieved (0)	Marks
System description is clearly written using LaTeX including system purpose and features	Student successfully prepares the system description using LaTeX and includes key details such as purpose and features	Student does not prepare the description in LaTeX or the description is incomplete	1
UML diagram correctly represents the selected system	Student creates a correct UML diagram (Use Case / Class / Sequence) representing the system functionality	Diagram is missing, incorrect, or does not represent the system properly	1
User Interface prototype designed using Figma	Student designs at least two UI screens in Figma demonstrating the main functionality of the system	UI prototype is missing or fewer than two screens are designed	1
Identification of design principle or design pattern	Student correctly identifies and briefly explains a suitable design principle or design pattern for the system	Design principle or pattern is missing or incorrectly explained	1
Overall understanding and integration of concepts from previous labs	Student demonstrates clear understanding and integrates concepts learned in previous lab sessions	Student shows weak understanding or fails to integrate previous lab concepts	1
Weighted CLO Score			
Remarks and Signature			

Software Description: RideNED

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Task 1: Software Description

System Name

RideNED

Purpose of the System

RideNED is a transportation management mobile application designed for NED University. The system aims to automate and digitize the university's transport operations. It bridges the gap between students and the transport administration by providing real-time point tracking, digital passes, and streamlined communication for complaints and lost/found items.

Main Features

- **Live Point Tracking:** Real-time GPS tracking of university points with ETA updates and stop notifications.
- **Digital Point Pass:** QR-based monthly passes linked to student accounts, replacing manual ID checks.
- **Digital Fee Payment:** Integration with E-wallets and Bank Transfers for seamless point fee payments.
- **Pre-Booking System:** Allows students to book seats for department-specific or general industrial trips.
- **Lost and Found Board:** A centralized forum to report and claim lost items on campus transport.

Types of Users

- **NED University Students:** Users who utilize the app to track points, generate passes, pay fees, and report complaints. Requires valid university credentials.
- **NED Admin:** Administrators who manage routes, approve digital passes, monitor live tracking, and handle driver complaints via a backend dashboard.

LaTeX Source Code used for Task 1

```
\documentclass{article}
\usepackage[utf8]{inputenc}
\usepackage{geometry}
\geometry{a4paper, margin=1in}
\title{\textbf{Software Description: RideNED}}
\author{Lab Session 08 - Open Ended Lab}
\date{}
\begin{document}
\maketitle

\section*{System Name}
RideNED

\section*{Purpose of the System}
RideNED is a transportation management mobile application
designed for NED University. The system aims to automate
and digitize the university’s transport operations.
It bridges the gap between students and the transport
administration by providing real-time point tracking,
digital passes, and streamlined communication for
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\section*{Main Features}
\begin{itemize}
\item \textbf{Live Point Tracking:} Real-time GPS tracking
of university points with ETA updates and stop notifications.
\item \textbf{Digital Point Pass:} QR-based monthly passes
linked to student accounts, replacing manual ID checks.
\item \textbf{Digital Fee Payment:} Integration with E-wallets
and Bank Transfers for seamless point fee payments.
\item \textbf{Pre-Booking System:} Allows students to book
seats for department-specific or general industrial trips.
\item \textbf{Lost and Found Board:} A centralized forum to
report and claim lost items on campus transport.
\end{itemize}
\section*{Types of Users}
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app to track points, generate passes, pay fees, and report
complaints. Requires valid university credentials.
\item \textbf{NED Admin:} Administrators who manage routes,
approve digital passes, monitor live tracking, and handle
driver complaints via a backend dashboard.
\end{itemize}

\end{document}
```
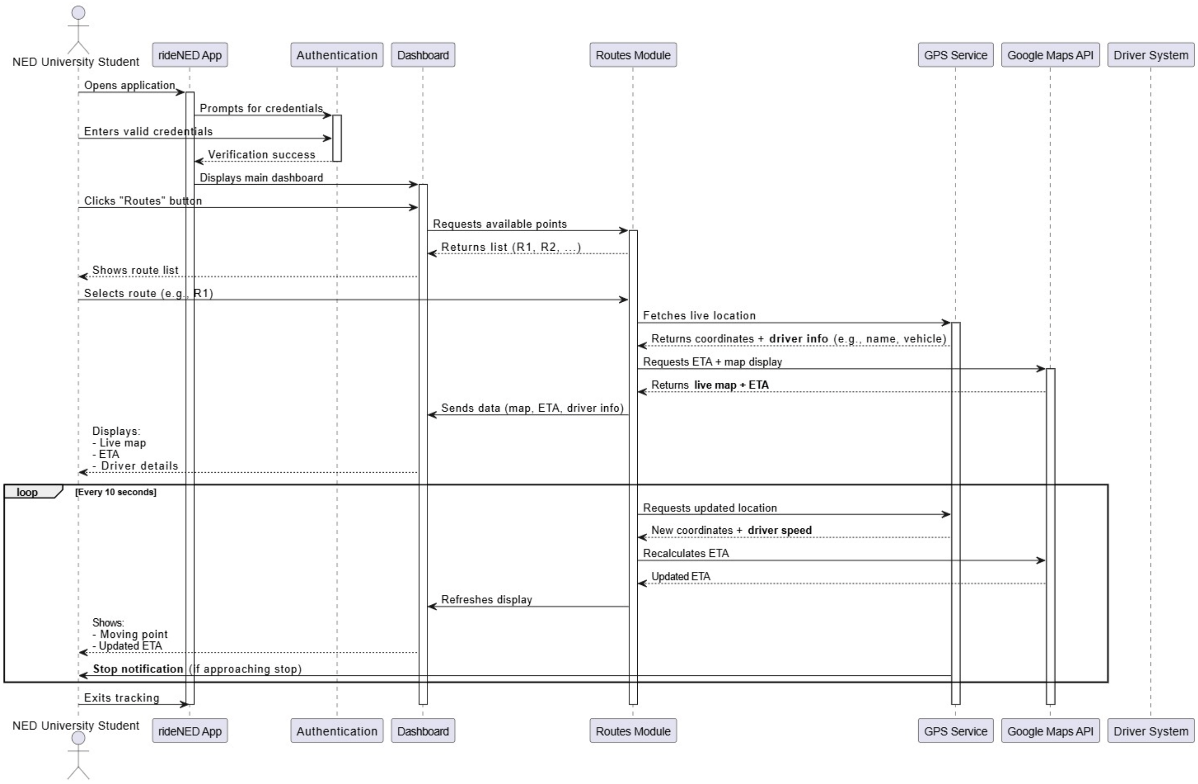
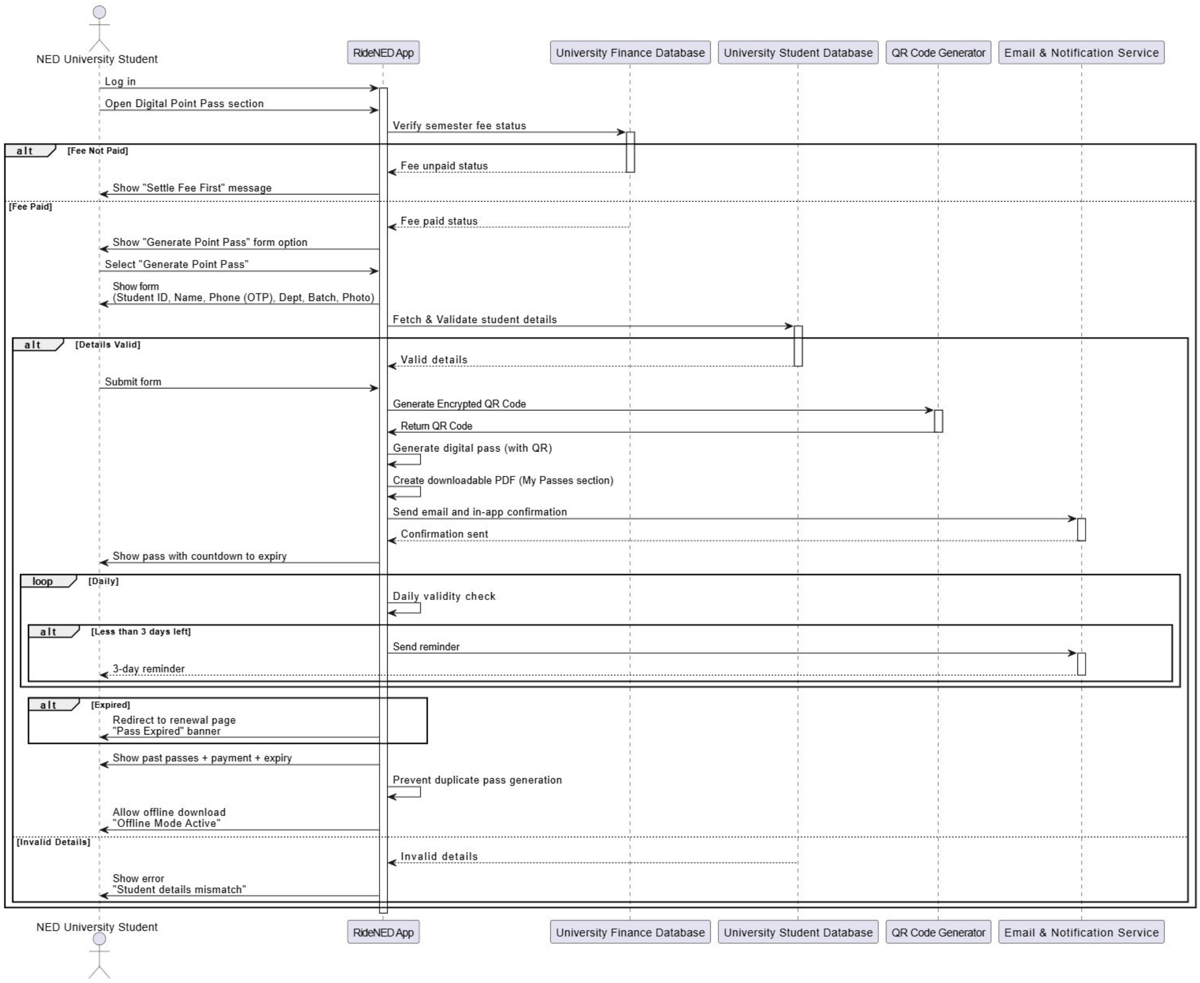

Task 2: UML Modeling

SEQUENCE DIAGRAMS

System Feature: University Point Tracking System



System Feature: Digital Point Pass



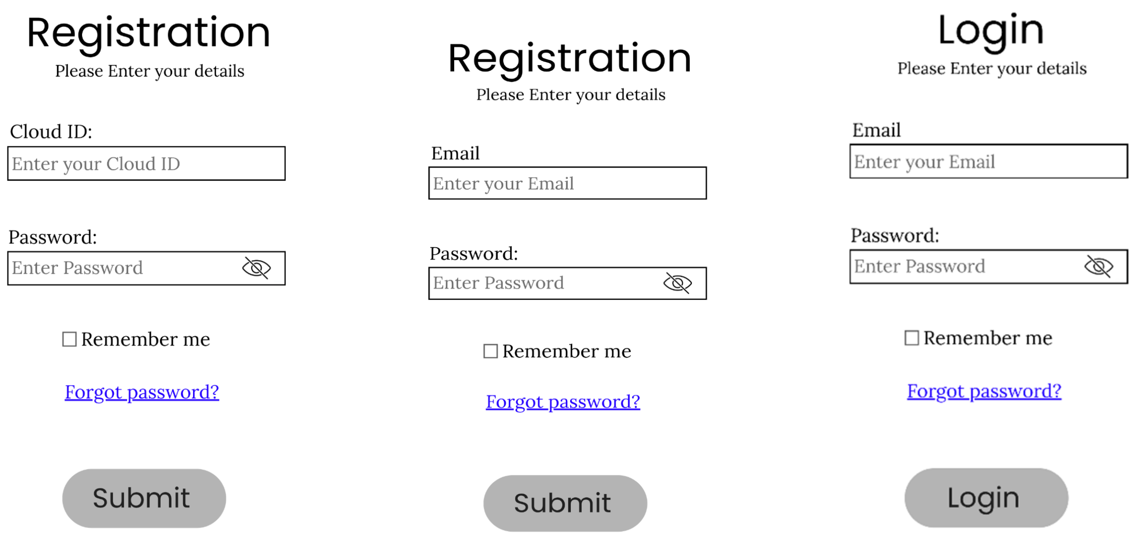
Task 3: User Interface Prototype

Registration Interface for Driver, Head of Department, and Admin

The registration interface allows the Driver, Head of Department (HOD), and Admin to create an account using their allocated credentials. The interface includes input fields for username and password, along with "Forgot Password" and "Remember Me" options to assist in credential recovery and user convenience.

Login Interface for Student, Driver, Head of Department, and Admin

The login interface provides secure access to the system for Students, Drivers, Heads of Department, and Admins. It contains two primary input fields, email and password, along with "Forgot Password" and "Remember Me" options to support account recovery and session management.



Dashboards Overview

Driver Dashboard

Driver Dashboard serves as the central hub for all driver-related operations. The interface includes options to:

- View active points
- Make announcements regarding delays or schedule changes
- Access Digital Pass for attendance
- Report or view lost and found items

- File complaints or request assistance
- View schedule

Driver can also view notifications, manage profile, and request assistance from the menu bar.

Admin Dashboard

The Admin Dashboard provides access to core administrative functions. Admins can:

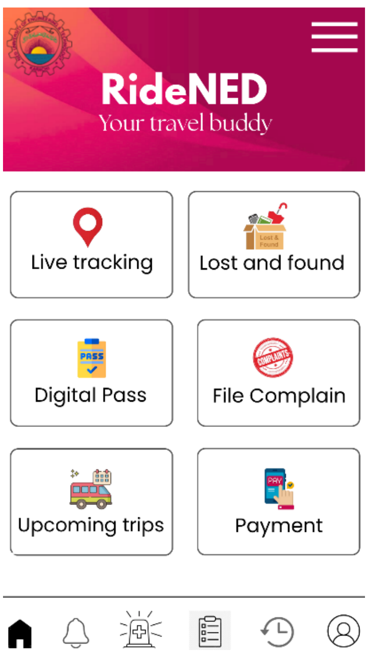
- View active points
- Handle complaints and feedback
- Book trips
- Make announcements visible to users
- View lost and found items
- Manage driver schedules

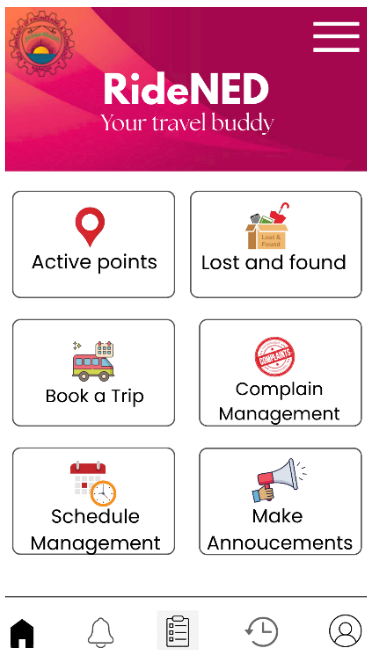
Admins can also view notifications, history, all digital passes, and manage their profile from the menu bar.

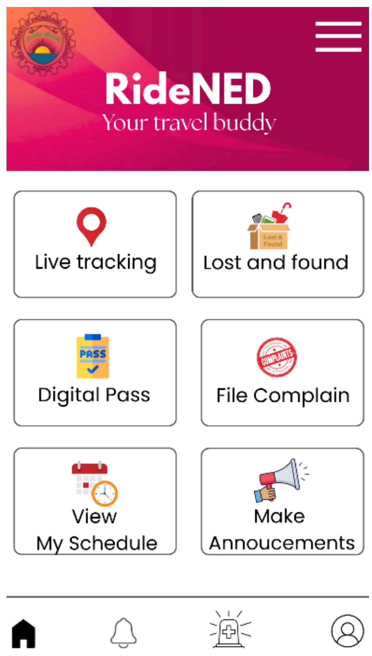
Student Dashboard

The Student Dashboard provides a visual overview of available services. Students can:

- Access the Digital Pass section to view or generate their pass
- Make payments for pass renewal or fees
- Report issues or emergencies
- View notifications and points lists
- Check history

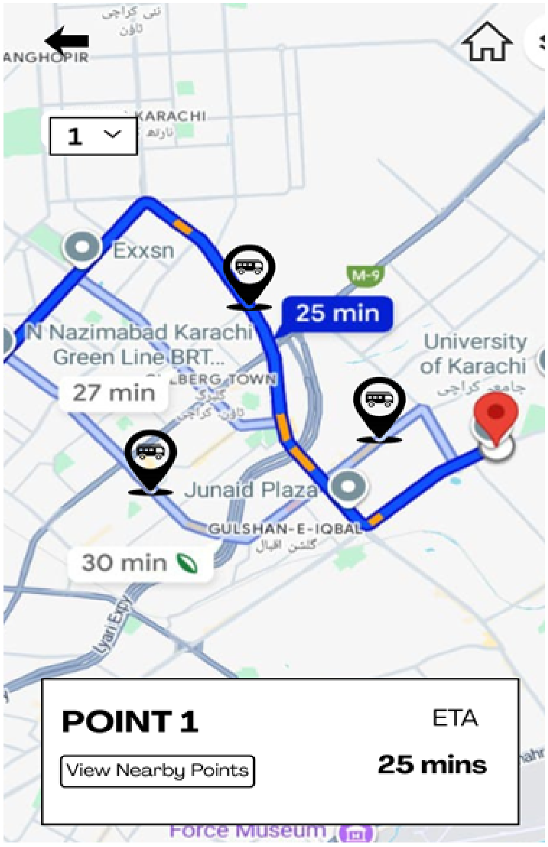






LIVE TRACKING FOR STUDENTS AND ADMIN INTERFACE

The live tracking interface provides a real-time map view that allows both students and administrators to monitor the movement of the point. Through this interface, users can track the point as it follows a selected route, view nearby points, and observe the estimated time of arrival (ETA). All tracking information is updated dynamically to ensure accurate and real-time visibility.



Task 4: Design Concept Identification

Design Pattern Identified: Observer Pattern

Possible Use in RideNED System:
The Observer Pattern is ideal for the Live Point Tracking and Notification System within the RideNED application.

Implementation Concept:

- The university point's GPS location service acts as the Subject (state owner).
- The students' mobile applications act as Observers (dependents).

How It Works:
Instead of having thousands of student apps constantly query the server for the bus location (polling), the Observer Pattern provides an efficient, event-driven solution.

- When a student clicks "Notify Me" for a specific stop, their app subscribes to that stop's route.
- The GPS tracker (Subject) updates its location state in real-time.
- If the estimated arrival time (ETA) is under 5 minutes, the system automatically sends a notification to all subscribed Observers.

Benefits:

- Decouples the GPS module from the user interface.
- Reduces server load and database queries.
- Ensures students receive real-time, accurate notifications efficiently.